

DecodeLabs Data Analytics Internship

Project 1: Data Cleaning & Preparation

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Internship: DecodeLabs Data Analytics
Internship

Project Overview

Data cleaning is a crucial step in the data analytics process. Raw datasets often contain missing values, inconsistencies, duplicate records, and formatting issues that can affect analysis results. The purpose of this project was to inspect, clean, and prepare a customer order dataset to ensure data quality and readiness for future analysis.

Objectives

The main objectives of this project were:

1. Identify missing values within the dataset.
 2. Handle incomplete data appropriately.
 3. Verify data consistency and accuracy.
 4. Check for duplicate records.
 5. Prepare a clean dataset for analysis.
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Dataset Description

The dataset contained customer order information, including:

- Order IDs
- Product Information
- Quantity Details
- Coupon Codes
- Customer Purchase Records

The dataset was reviewed to identify potential data quality issues before analysis.

Methodology

The following steps were performed during the data cleaning process:

1. Data Inspection

The dataset was opened and examined to understand its structure, columns, and data types. Each column was reviewed for missing values and inconsistencies.

2. Missing Value Analysis

The CouponCode column contained missing values. Missing values can create problems during analysis and reporting if left unresolved.

3. Missing Value Handling

All missing values in the CouponCode column were replaced with:

NO_COUPON

This value indicates that no promotional coupon was used during the corresponding transaction.

4. Duplicate Record Verification

The dataset was checked for duplicate records to ensure data accuracy and prevent repeated entries from affecting future analysis.

Result:

- No duplicate records were found.
- No duplicate Order IDs were detected.

5. Data Validation

The dataset was reviewed to verify:

- Consistent formatting of records
 - Valid numerical values
 - Proper structure across all columns
 - Overall data quality and integrity
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Results

After completing the data cleaning process, the following results were achieved:

Before Cleaning

- Missing values existed in the CouponCode column.

After Cleaning

- Missing values successfully resolved.
 - CouponCode column standardized using "NO_COUPON".
 - No duplicate records identified.
 - Dataset prepared for analysis.
 - Data quality improved and validated.
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Learning Outcomes

Through this project, the following skills were developed:

- Data Cleaning Techniques
 - Missing Value Handling
 - Data Validation
 - Dataset Inspection
 - Data Preparation for Analytics
 - Documentation and Reporting
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Conclusion

This project successfully demonstrated the fundamental processes involved in data cleaning and preparation. By identifying and resolving missing values and validating the dataset structure, the data was transformed into a clean and analysis-ready format.

Data cleaning is an essential step in the analytics workflow because accurate and reliable insights depend on high-quality data. This project provided practical experience in preparing datasets for future analytical tasks and strengthened foundational data analytics skills.

Tools Used

Microsoft Excel / WPS Spreadsheet

GitHub

Microsoft Word

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